

Reference Definitions Reverse Detector Rankings in Streaming Key Estimation

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Abstract. A streaming key estimator must predict after each event without future input and may abstain when its evidence is insufficient. Its score also depends on what the reference means by “key.” We define a selective-prediction protocol measuring coverage, claimed-event accuracy, stability, and key-change lag, and use two frozen causal detector packages to test reference sensitivity. On 36 Beethoven performances, we hold the inputs, detector outputs, common claimed events, and a 12-class diatonic-collection ontology fixed. Analyst-declared key contexts and active notated key-signature collections have mean per-piece agreement 0.65 and reverse the package ranking: the long-memory minus short-memory accuracy difference changes from -0.081 to $+0.080$ (exploratory difference of differences $+0.160$, CI95 [$+0.118$, $+0.205$]). Development analyses support the consequence for model selection: independently crossing evidence memory and a harmonic-function term produces opposite accuracy effects across When in Rome and Isophonics reference regimes, with coverage reported alongside accuracy. Restricting the same performed-input analysis to longer-persistence analyst-key segments also shifts relative accuracy toward the long-memory package, without establishing a sharp threshold. These results do not make one reference uniquely correct or isolate temporal granularity from repertoire in the cross-corpus evidence. They show that reference provenance, semantics, persistence, and tonal ontology are part of the task definition and must accompany streaming key-estimation scores.

1 Introduction

A musician playing a MIDI keyboard motivates a causal key indicator: after each recognized chord, the estimator must name a key or withhold judgment, without a second pass or future context. Incremental tonal interpretation is not new. Weber’s analysis already treated a listener’s tonal understanding as something revised while events unfold [1], [2], and modern perceptual and interactive systems have studied real-time tonality induction [3], [4]. Retrospective and in-time readings are both valid; they answer different analytical questions. This paper studies the narrower computational case of causal streaming inference from chord-recognition output.

That information boundary changes the evaluation. The observations are uncertain symbolic events: ranked chord identities, voicings, timing, and hold duration produced by an upstream recognizer. The detector may abstain, so coverage and accuracy on claims must be read together. It may also change its answer, making stability, key-change matching, lag, and

time-to-first-claim relevant. These requirements motivate the protocol in Section 3, but they do not settle a more basic question: what reference makes a prediction correct?

The corpora used here supply three different constructs. When in Rome gives an analyst-declared key context inside a functional analysis. Isophonics supplies time-aligned tonality-region annotations. ASAP supplies active notated key signatures, which identify a major/relative-minor collection rather than a unique tonic and mode. Their temporal persistence differs, but so do their provenance, semantics, repertoire, and tonal assumptions. Calling them merely “local” and “section” keys hides those differences.

The central experiment holds the performed inputs and both detector outputs fixed, maps predictions and references to the same 12 diatonic collections, and changes only whether scoring follows the analyst context or the notated key signature. The reference choice reverses which frozen detector package scores better (Section 6). This result is not the familiar observation that a long window is smoother. It shows empirically that two documented notions of key can select different systems on unchanged predictions.

The contributions are therefore ordered as follows. First, a controlled reference-construct analysis demonstrates the ranking reversal and quantifies the disagreement between the references. Second, independent memory/function crosses and a piece-aware segment-persistence analysis show how the evaluation regime changes model selection without attributing every cross-corpus effect to timescale alone (Section 6). Third, a frozen causal HMM family and a selective-prediction protocol provide an interpretable experimental instrument, held-out checks, and reproducible artifacts (Section 5, Section 9). The HMM architecture, the idea of in-time analysis, and the classic offline reference analyzers are not claimed as novel.

2 Related work

Distributional key finding begins with the Krumhansl-Schmuckler probe-tone profiles [5] and revisions such as Temperley’s [6], with corpus-trained profiles improving minor-mode behavior [7]. Temporal structure enters through Bayesian and hidden Markov treatments [8], [9]. The closest architectural analogue here is justkeydding [10], an HMM over profile emissions for global and local key. Neural symbolic-harmony systems also estimate local key within larger retrospective analyses [11], [12]. Our HMM remains deliberately close to this interpretable lineage.

Global key conventionally characterizes a whole song, piece, or movement; local-key estimation emits finer-grained predictions [13], [14]. That output distinction is independent of the information boundary: either task may be causal or retrospective. Nor does a local-key sequence uniquely encode music-theoretical modulations and tonicizations [14]. The reference must therefore be described operationally rather than inferred from the word “local.”

Sequential key induction has a substantial literature. Rowe reviewed key finders for interactive performance that operate as material arrives [3]; Toiviainen and Krumhansl modeled continuously developing listener judgments [4]; and Chuan and Chew presented a real-time polyphonic-audio key finder [15]. Listener experiments likewise show that perceived modulation depends on the preceding musical context [16]. These precedents make causality an established task choice, not this paper’s novelty. This study instead contributes selective prediction from chord-recognition events and a controlled analysis of reference sensitivity.

Reference design is especially consequential across repertoires. The major/minor ontology fits much common-practice analysis but does not exhaust popular tonal organization. The repeated Am-F-C-G axis progression can support both A-minor/Aeolian and C-major hearings [17], and experiments on rotations of the same four chords show that chord identity and metric placement both influence perceived center [18]. Six-based minor and other major/minor mixtures further resist a forced binary reading [19]. Abstention can express insufficient evidence among modeled labels; it cannot represent a confident modal, blues-based, or dual-tonic answer outside the 24-state space.

Evaluation follows the MIREX weighted score for near-key errors [20], applied only within our corpora. The reference sources are not interchangeable. When in Rome is a meta-corpus of human functional analyses encoded in RomanText [21], [22]. Isophonics provides time-aligned tonality regions, normalized through ChoCo [23], [24]. ASAP provides score and performance alignment plus key-signature metadata [25]. Section 4 traces how each source is transformed and what agreement with it can mean.

Bayesian online changepoint detection (BOCPD) [26] is evaluated as an adaptive-memory alternative to fixed decay. Recent autoregressive extensions address within-regime dynamics [27]; our categorical implementation is a secondary model analysis rather than part of the central contribution.

3 Task and evaluation protocol

Task. The input is a causal stream of chord events from the application’s capture path: each event carries ranked chord candidates with explanation costs, the sounding voicing, the analysis-time tonality context, a timestamp, and a hold duration. An event is one held chord, not one keystroke: the capture

path aggregates pedal-aware sounding notes and commits an event only after its chord identity persists past a debounce and minimum-duration gate, so finger rolls and passing voicings are absorbed upstream of the detector. After each event the detector outputs either a ranked list of (key, confidence) hypotheses or an explicit abstention. It never sees the future and is never re-run on edited history. The key space is the 24 major and minor keys over pitch classes; enharmonic spelling is a presentation concern. This ontology is a modeling choice, not a universal account of tonality. A source annotation outside it is unscorable for 24-key accuracy rather than evidence that the detector should have abstained.

Metrics. The top-ranked claim is scored. Accuracy-like metrics are reported in the selective-prediction frame [28]: coverage (how often the detector makes a claim) and accuracy on claimed events form an inseparable pair, and abstentions are never counted as errors. Exact accuracy is complemented by the MIREX-weighted score, which grants partial credit for musically close misses (fifth 0.5, relative 0.3, parallel 0.2) [20]. At zero coverage, however, accuracy on claims is undefined rather than perfect, so an always-abstaining system cannot win the evaluation.

Temporal behavior gets three measurements. An annotated key change is **matched** when the detector claims the new key before the next change, and its **lag** is counted in events; unmatched changes are reported as censored counts, never averaged into lag. A switch between claims is **spurious** only when the annotation did not change and the new claim is not the annotated key, so a lagged catch-up onto the right key is never penalized twice. Its per-piece distribution includes only pieces with at least one exact local-key reference; raw switches and time to first claim remain behavioral summaries over every piece. Annotated changes require adjacent non-null references, so null-reference regions enter neither the matched-change count nor the lag denominator. **Time-to-first-claim** counts events before the detector first commits. These counts are long-tailed, so each is summarized by the per-piece median and 90th percentile (p90) rather than a mean. Ambiguity-labeled events (hand-authored fixtures only) accept abstention or any acceptable key.

Two diagnostics complete the picture. Selective-prediction behavior is the coverage-accuracy curve swept over the abstention threshold, with the selected operating point marked. For probabilistic detectors we also report top-label posterior reliability: events are binned by the posterior probability of the leading key (ten equal-width bins) and compared with exact correctness, alongside expected calibration error (ECE), negative log likelihood, and Brier score. Reliability is separate from abstention: a detector can abstain informatively while still being overconfident in its posterior probabilities.

Statistics. Every adoption decision compares two configurations piece by piece, using the Wilcoxon signed-rank test and

corpus	repertoire / input	reference	pieces / events
When in Rome [21]	common-practice / score-derived	analyst-declared key context	77 / 5,207
ASAP [25]	classical piano / performed MIDI	active key-signature collection	60 / 19,546
Isophonics [23], [24]	popular songs / synthesized voicings	time-aligned tonality region	224 / 19,062
ASAP-WiR overlap	Beethoven / performed MIDI	analyst context and signature collection	36 / 10,395

Table 1: Evaluation sets and their reference constructs. Repertoire, observation construction, and reference provenance change together in cross-corpus comparisons.

a seeded-bootstrap 95% confidence interval on the mean per-piece difference. The piece is the unit of analysis because events within a piece are strongly dependent and a few long pieces dominate pooled counts; pooled event accuracy is never decisive. The signed-rank test assumes no distributional form for these small samples. P-values on development data served model selection and are uncorrected for multiplicity; confirmatory weight rests on the original predeclared held-out result package (Section 9). Bootstrap intervals added in the revision analyses are labeled exploratory or descriptive according to their protocols.

Protocol discipline. The protocol was frozen before detector tuning, with changes recorded as dated amendments. Development/test splits were frozen per corpus before the first experiment on it (by piece, and by composer where the corpus allows); all original tuning, ablation, and model selection ran on development splits. The paper-era held-out result package was declared before its single execution. After editorial screening, four revision analyses were declared in a dated protocol before execution: a scorable-cohort correction, a piece-aware segment reanalysis, a dual-reference sensitivity analysis, and a development-only memory/function grid. They do not tune a new system or promote post-hoc results into the original confirmatory record. The evaluation harness structurally strips labels before events reach a detector.

4 Data and reproducibility

Evaluation uses **fixtures**: versioned event streams containing the chord recognizer’s candidate rankings, observed voicings, timing metadata, and a parallel reference stream read only by the scorer. Fixtures are generated under a fixed neutral analysis context: the recognizer never sees a reference key, so its tonality-sensitive ranking rules cannot leak the answer into the observations. Results are comparable only within a fixture version because each fixture embeds a specific frozen chord-recognition profile.

Table 1 separates repertoire, observation construction, and reference provenance. The first three sets have frozen development/test splits. The ASAP-WiR overlap is evaluation-only and supports both performed-input analyst contexts and, in the revision analysis, the active ASAP key-signature collection.

The When in Rome subset contains 77 common-practice works by Bach, Brahms, Schubert, Beethoven, and Mozart. Its Roman-Text key field is the analyst’s current interpretive context; an

applied-chord figure can mark a tonicization without changing that field. The fixtures contain 514 explicit key changes and 455 events with applied-chord slash figures, so “analyst context” must not be read as “every tonicization.”

The Isophonics set contains 224 popular recordings, mostly the Beatles, with Queen, Zweieck, and Carole King also represented; all 41 held-out tracks are Beatles recordings. The source describes its keys as time-aligned tonality regions, not uniformly whole-song labels, and notes that changes may be omitted. Plain major/minor regions map into the detector’s ontology. Seven tracks contain 568 modal or no-key events outside it; three held-out tracks are wholly modal. They remain in the frozen split but are excluded from both accuracy and coverage in the corrected 38-track 24-key cohort (Section 9).

ASAP key signatures are not analyst-stated local keys. At each event the signature defines an acceptable major/relative-minor pair; it cannot identify a unique tonic and mode and may remain notationally unchanged across an analytical modulation. For the dual-reference analysis, both signature pairs and major/minor analyst contexts are mapped to the corresponding 12 diatonic- collection classes.

The overlap transfers When in Rome analyst contexts onto ASAP performances of the same Beethoven sonata movements through score-performance downbeat alignment. Its per-movement measure offset is calibrated by content: agreement between sounding pitches and analyst chords must peak sharply at zero when labels are slid by whole measures. The project corrected erroneous offsets in 11 of 36 movements before peer review; 347 of 10,395 event labels changed, and all reported conclusions retained their direction.

Performed fixtures are created by offline replay of recorded MIDI with sustain semantics through the production chord analyzer, three-note gate, and event- segmentation core. The detector later consumes the stored events in causal timestamp order. This exercises the core musical-analysis path on performed input; it is not an end-to-end live-user, transport, latency, display, or usability evaluation.

A small hand-authored pop/jazz suite supplies behavioral regressions outside all pooled statistics. The repository contains the open fixtures, split files, evaluation harness, paired-comparison code, dated logs, and frozen result artifacts. License-gated fixtures remain in local builds generated from pinned upstream checkouts. Named paper-era analysis profiles and

detector recipes reproduce the original fixtures and all nine frozen result directories even though later application defaults have changed.

5 Model family and frozen packages

The detector family is intentionally interpretable: profile matches, key-relative chord-function scores, and a decaying progression score, with no learned embeddings. Its floor is a duration-weighted, exponentially decaying pitch-class histogram ranked against all 24 rotations of the corpus-trained Albrecht-Shanahan major/minor profiles [7]. A functional term scores diatonic membership, dominant function, and leading-tone function; the broader progression term scores a decaying history of several cadential patterns.

Two frozen packages organize the principal comparisons. The **long-memory package** uses 30-second profile emissions with functional and progression blends at zero. The **short-memory package** uses 1-second emissions plus functional blend 0.1; its progression blend is also zero. Both use the same HMM, duration weighting, abstention rule, and same-tonic mode cue. These packages were selected under different development reference regimes before the original held-out evaluation. They are experimental configurations, not current application defaults.

The detector family is a causal HMM over the 24 major and minor keys. It keeps a filtered posterior and updates it by the forward algorithm only; no Viterbi decoding or future context is used. At each event, the update is:

```
for each pitch class p:
    histogram[p] =
        sum over prior events i of:
            duration_i
            * 2^(-age_i / half_life)
            * indicator(p in voicing_i)

for each candidate key:
    profile_score[key] =
        correlation(histogram, key_profile[key])

emission = softmax(profile_score / temperature)
prior =
    transpose(transition) * previous_posterior
posterior = normalize(emission * prior)
```

Multiplication in the last line is elementwise. In words: the profile scores become an event-level likelihood over keys, the previous belief is advanced through the transition model to form a prior, and the two are combined and renormalized into the filtered belief after the current event.

In the long-memory configuration, `key_profile` is the Albrecht-Shanahan major/minor profile set [7]. Each event’s contribution to the histogram is multiplied by its duration, and `half_life = 30 s` means that contribution halves after 30 seconds. The softmax temperature is 0.25, sharpening the

profile-correlation scores into a more selective likelihood. The transition matrix assigns 0.9 probability to remaining in the same key; the remaining mass is distributed over other keys with decay by circle-of-fifths signature distance. The frozen detector withholds its first two events. From the third event onward, it reports the posterior’s top key and probability only when the margin between the top two probabilities is at least 0.3; otherwise it abstains. The three-event warm-up and margin floor are independent claim gates.

Two modeling constraints matter for interpreting the results. First, the decayed histogram and the HMM’s state persistence are both memories, and evidence entering one is carried forward by the other. An early configuration fed the HMM emission scores that already integrated 30 s of history and expected the transitions to supply key tracking on top; the posterior saturated and lagged behind corrections. The two mechanisms therefore hold distinct jobs: the histogram defines what one observation is, and the transitions supply persistence across hidden keys. Second, the emission memory controls how local each observation is. The next section measures how its value, and the value of functional evidence, changes across explicitly defined reference regimes.

6 Reference-dependent evaluation

6.1 Fixed outputs, different references

The primary analysis asks whether two documented reference constructs select the same detector on unchanged material. It uses all 36 ASAP-WiR Beethoven performances and the frozen long- and short-memory claim streams. Analyst keys, key signatures, and detector claims are mapped to 12 diatonic-collection classes: a minor key shares a class with its relative major. This gives both references the same cardinality while deliberately discarding major/minor identity. The primary event mask contains the 8,160 events on which both packages claim and both references exist.

reference	long	short	long - short, CI95
analyst key context	0.581	0.661	-0.081, [-0.137, -0.026]
key-signature collection	0.626	0.547	+0.080, [+0.029, +0.128]

Table 2: Exact accuracy on common claims under two references, macro-averaged by performance in a shared 12-class ontology. Intervals are exploratory paired bootstrap intervals. Neither detector output changes.

Mean per-piece reference agreement is only 0.6465. Under the analyst context, the short-memory package leads; under the active key-signature collection, the long-memory package leads (Table 2). The piece-level difference of differences is +0.1602, with exploratory CI95 [+0.1184, +0.2046]; 31 of 36 pieces have a positive interaction. The same reversal appears when each package is evaluated on its own claims, so the common-event restriction does not create it.

This is a reference-construct sensitivity result, not a contest to identify the one correct reference. An analyst’s current key context and the collection implied by a notated signature answer different valid questions. The result is also not a pure timescale manipulation: source, semantics, and persistence all change with the reference. What the design establishes directly is that the reference definition can reverse model selection while performances and predictions remain fixed.

6.2 Memory and function across references

The two frozen packages bundle memory and functional evidence, so their reversal alone cannot say which ingredient matters. A predeclared revision grid crosses half-life {1, 30} seconds with functional blend {0, 0.1} on both development sets while fixing every other paper-era setting (Table 3). It is exploratory mechanism evidence: the corpora still differ in repertoire, observation construction, and reference practice.

reference regime	func.	1 s	30 s
WiR contexts	0.0	0.546 / 0.680	0.434 / 0.784
WiR contexts	0.1	0.601 / 0.764	0.514 / 0.824
Isophonics regions	0.0	0.736 / 0.791	0.775 / 0.921
Isophonics regions	0.1	0.629 / 0.815	0.713 / 0.926

Table 3: Development cell means when memory and functional evidence are crossed independently; detector cells report exact accuracy / coverage. Exact accuracy is conditional on claims; WiR is When in Rome.

On When in Rome, 30-second minus 1-second exact accuracy is -0.104 without functional evidence and -0.089 with it; both exploratory intervals exclude zero. On Isophonics the corresponding effects are $+0.040$ (CI95 $[-0.0001, +0.0786]$) and $+0.084$ (CI95 $[+0.041, +0.129]$). Function helps When in Rome at either memory ($+0.063$ and $+0.081$) and harms Isophonics at either memory (-0.107 and -0.062); all four functional-effect intervals exclude zero. Longer memory raises coverage in every pair, while functional evidence raises or preserves it, so the opposite exact-accuracy effects are not produced by opposite abstention directions.

The familiar smoothing explanation correctly anticipates that persistent references may reward longer evidence windows. It does not by itself predict the magnitude of the reference disagreement in Table 2 or the sign change of a modeling ingredient in Table 3. Those are the evaluation and model-selection consequences established here.

Figure 1 sweeps the emission-memory half-life from 1 to 60 seconds on both development corpora. The plotted series confirm the endpoint pattern without being strictly monotonic: both When in Rome curves reach a minimum at 15 s and partially rebound, while Isophonics reaches a plateau by 8 s. The R4 endpoint contrasts above supply the paired uncertainty for newly emphasized inference; the full inspected sweep is descriptive.

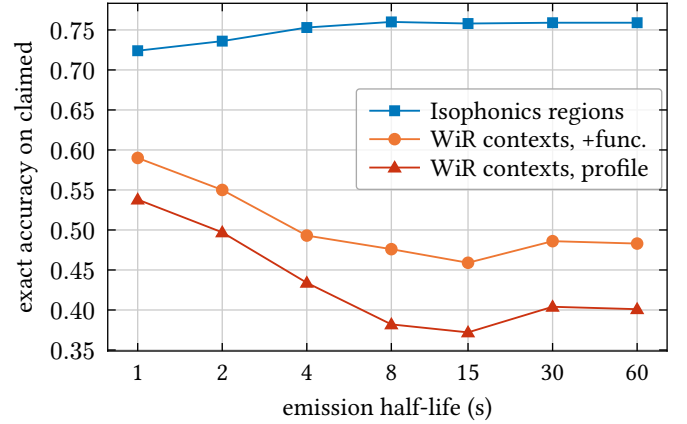


Figure 1: Descriptive development sweep. When in Rome exact accuracy declines strongly to 15 s and partially rebounds; Isophonics rises to a broad plateau from 8 s. Coverage rises with memory in all three series. No one memory setting is favored by both reference regimes.

6.3 Reference persistence on performed input

The overlap also permits a within-reference descriptive analysis. Each event keeps its projected analyst context; the analysis only restricts eligibility to contexts that persist for at least 0, 12, 20, or 32 score measures. It therefore changes the event subset, not the labels assigned to identical events. The thresholds and pooled direction had already been inspected, so Table 4 uses piece-level summaries and descriptive intervals rather than a new confirmatory threshold claim.

min. span	n	long	short	diff. / CI95
0	36	0.895 / 0.502	0.846 / 0.605	-0.103, [-0.158, -0.046]
12	36	0.909 / 0.600	0.863 / 0.628	-0.028, [-0.084, +0.029]
20	35	0.907 / 0.625	0.864 / 0.624	+0.001, [-0.067, +0.069]
32	30	0.913 / 0.691	0.875 / 0.644	+0.046, [-0.030, +0.120]

Table 4: Piece-level ASAP-WiR results as evaluation is restricted to longer- persistence analyst-key segments. Detector cells report coverage / exact accuracy; spans are score measures. Intervals are descriptive paired bootstrap intervals. Contributing pieces and coverage change with the filter.

Relative accuracy shifts progressively from the short-memory package toward the long-memory package as the minimum segment span rises. The same trend remains on the common-claim events: long-minus-short exact accuracy moves from -0.093 to $+0.033$. Only the all-event short-memory advantage has an interval excluding zero; the later near tie and long-memory lead are descriptive. This supports an association between reference persistence and package fit, not a sharp 12-

measure crossover or proof that persistence alone explains the cross-corpus results.

Together, the three analyses establish different levels of evidence. The dual-reference result directly holds predictions fixed. The development grid shows that package bundling does not create the memory and functional patterns. The persistence analysis connects those patterns to a measured temporal property within one performed-input analyst reference. Repertoire and observation construction remain alternative contributors to the broader When in Rome versus Isophonics contrast.

6.4 Secondary ablations and selective prediction

The original factorial also tested the decaying progression-score emission blend, duration weighting, and recognizer-confidence weighting. The progression blend did not improve the HMM cells; duration weighting improved both principal development regimes (+0.02 to +0.05 exact) and was retained; recognizer- confidence weighting was inert in five paired comparisons. These verdicts apply to the mechanisms tested, not to all uses of harmonic motion or recognizer uncertainty. In particular, post-submission product work adopted a narrower cadence-conditioned transition mechanism; it does not reverse the frozen negative for the broader emission-side progression score and is not part of the paper’s result package.

Sweeping the posterior-margin floor traces the selective-prediction curve in Figure 2. Higher thresholds reduce coverage while raising accuracy on the remaining claims. An all-abstaining detector would have zero coverage and no claimed-event accuracy; it cannot score as a useful system. The margin floor of 0.3 was selected on development data before held-out execution. That point is evaluated as a coverage-accuracy pair, with stability and latency reported separately.

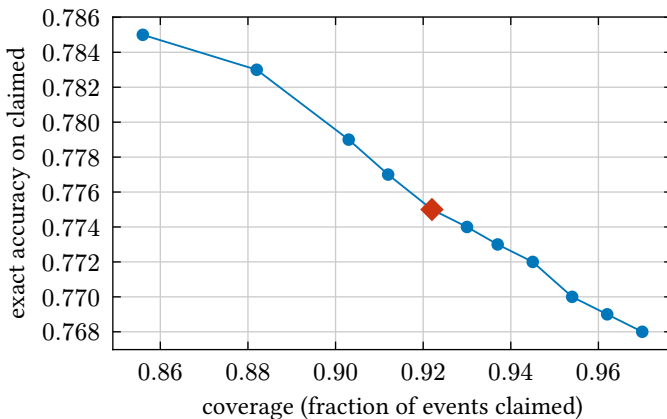


Figure 2: Selective-prediction behavior of the frozen long-memory package on the Isophonics development split, swept over the posterior-margin floor (0 to 0.6). Moving left raises the margin required to speak, so coverage falls; moving up means the remaining claims are more often correct. The marked point is the evaluated operating point (floor 0.3).

7 Mode disambiguation

On mode-resolved corpora the residual errors sort into MIREX classes: fifth-neighbor confusions, relative-pair confusions (shared signature), and parallel-pair confusions (shared tonic). Mode errors, the most user-visible class since an indicator displays major or minor, were about 10% of claims on both mode-resolved development regimes after fixing a detector package.

The design constraint that works is mass preservation within a musically related pair: if a cue only redistributes evidence between two hypotheses that share an invariant, it can help choose between them while leaving every other key untouched.

Parallel pairs (selected). Parallel keys share the tonic pitch class, and practical minor borrows the raised sixth and seventh, so the discriminating evidence collapses to roughly one pitch class, easily outvoted in a 12-dimensional correlation. Symbolic observations change that: when the played chord is rooted on a candidate tonic and its quality is one a tonic chord can carry (plainly major or minor), that quality is the most direct mode cue available. The cue is applied as a log-odds tilt within the parallel pair after the emission softmax, then rescaled so the pair’s total emission is unchanged. By construction it cannot move tonic evidence at all, which rules out the failure mode of two less constrained ancestors, an unconditional bonus for tonic chords and the functional rules of Section 5, both of which could fight modulation tracking. Strength 2 was selected after a 0.25-to-4 sweep with a broad 2-to-4 plateau: paired exact wins in both development regimes (Isophonics +0.016, $p = 0.030$; When in Rome +0.030, $p = 0.029$), parallel confusion roughly halved everywhere measured (4% to 2% of claims on Isophonics). Across the Isophonics sweep, coverage varied within 1.2 percentage points, spurious p90 remained 1, and matched annotated changes rose from 82 to 102 of 192.

Relative pairs (measured negative). The same pattern generalizes with the key signature as the conserved quantity. But here the evidence is structurally weaker. In a parallel pair, hearing the rival’s tonic chord is rare mode mixture; in a relative pair it is ordinary diatonic harmony (A minor’s tonic triad is just the vi chord of C major), so isolated chord quality fires constantly for the wrong twin. Two sharpened cues were built and swept: a bass-gated tilt (fires only when the chord’s root is also its bass) and a cadential-bigram tilt (a dominant-quality chord resolving down a fifth onto a home-quality chord). This particular cadence cue is inert. The bass cue works mechanically (relative confusion 6% to 4% of claims) but misses paired significance on Isophonics (+0.007, $p = 0.055$), trends negative on the other (-0.009, $p = 0.056$), and doubles spurious p90; unlike the parallel tilt’s broad plateau, it turns harmful at strength 2, the signature of a weak signal. Not adopted; retained in code as a measured negative.

Fifth pairs (not adopted). Fifth neighbors conserve neither tonic nor signature, and every key has two of them, so there is no pair to redistribute within. Discriminating a key from its dominant is the modulation problem, and any chord-quality nudge between fifth neighbors recreates the unconditional tonic bonus rejected above. Empirically the residual fifth errors also lack an exploitable shape, splitting nearly evenly between the dominant and subdominant sides (57/43 and 56/44 on the two mode-resolved corpora).

One pattern explains all three outcomes: a tilt is worth what its invariant is worth. Parallel keys share a tonic, and the discriminating evidence is rare and decisive, so the cue was adopted. Relative keys share only a signature, and the discriminating evidence is everyday harmony, so the cue was too weak. Fifth neighbors share nothing, so there is no pair to tilt within at all.

8 Adaptive temporal models

Explicit-duration modeling (HSMM). The HMM’s fixed self-transition implies geometric key-dwell times: at every event the key survives with the same probability. An HSMM (hidden semi-Markov model) would replace that with a richer dwell-time family. Before building one we measured whether the system even responds to the parameter an HSMM would refine: sweeping the self-transition from 0.7 to 0.98 (mean dwell roughly 3 to 50 events) moves every metric along the coverage-accuracy curve rather than off it (Isophonics exact within 0.008, spurious p90 1 through 0.95 and 2 at 0.98). The probe does not prove explicit- duration modeling useless for key estimation; it shows there is little headroom at the selected long-memory Isophonics operating point.

Bayesian online changepoint detection, built and measured. The dwell probe addresses the changepoint prior (constant-hazard BOCPD implies the same geometric dwell) but not BOCPD’s distinct contribution, the adaptive evidence window: instead of decaying evidence through a fixed 30 s half-life, BOCPD infers where the current section started and pools evidence back to that point. We built the constant-hazard detector for the 24-key space [26] with emission conventions identical to the frozen long-memory HMM (including the mode tilt), so any difference is attributable to the window (Table 5).

config	cov	exact	matched changes	spur med/p90
HMM, long	0.92	0.775	94/192	0/1
BOCPD h=1/200	0.88	0.715	161/192	5/14
BOCPD T=0.5	0.89	0.765	135/192	2/7
BOCPD T=1.0	0.90	0.769	115/192	0/4

Table 5: BOCPD versus the frozen long-memory HMM on the Isophonics development set. Here h is the constant changepoint hazard and T is emission temperature. The adaptive window matches more annotated changes, but no tuning recovers the long-memory stability point.

The adaptive window increases annotated-change matching by 70% because evidence resets at inferred starts. The same reactivity degrades stability under the Isophonics reference, and softening the per-event evidence does not recover the long-memory point (best cell: exact wash, $p = 0.51$, spurious p90 4 versus 1). At matched reactivity BOCPD does outperform the HMM’s fast setting (0.765 exact at 135 matched changes versus 0.736 at 141 for the HMM at a 2 s half-life), so adaptive windowing can improve responsiveness without being preferred by this reference regime. Its false alarms occur within stable annotated regions, where the independent-given-key assumption is weakest. Autoregressive extensions target analogous within-regime dependence in continuous series [27], but those Gaussian remedies have no direct analog for categorical chord emissions; the broader emission-side progression score tested here did not repair the limitation.

9 Held-out evaluation

The original paper-era held-out manifest was declared before execution and then frozen: it evaluates the long-memory package on all three splits, the short-memory package on the two mode-resolved splits, three music21 [29] profile-correlation analyzers on Isophonics, and a descriptive mode-confusion breakdown. Later product investigations reused some held-out pieces, so they are not independent new tests; no later result is used here. The original artifacts and the declared R1 correction are preserved with the project.

split	scorable / total	coverage	exact
Isophonics	38 / 41	0.884	0.732
WiR	18 / 18	0.811	0.587
ASAP	10 / 10	0.830	0.683

Table 6: The frozen long-memory package on the three held-out splits. Three wholly modal Isophonics tracks have no event in the 24-state ontology and are excluded from both accuracy and coverage, while remaining in the frozen split. ASAP is scored against acceptable major/minor realizations of its key signatures.

The long-memory package retains substantial coverage across all three inputs (Table 6). Development-to-test changes differ by corpus (Isophonics exact falls from 0.775 to 0.732; When in Rome rises from 0.434 to 0.587), so these small splits support a generalization check rather than a precise estimate of deployment performance.

The predeclared package ordering reverses across references. Against When in Rome analyst contexts, the short-memory package exceeds the long-memory package (0.649 versus 0.587 exact; paired difference +0.062, CI95 [+0.004, +0.121], $p = 0.047$; 16 paired pieces), at lower coverage (0.745 versus 0.811). Against Isophonics region labels, the long-memory package exceeds the short-memory package (0.732 versus 0.556; +0.175, CI95 [+0.040, +0.315], $p = 0.039$; 38 pieces), at higher coverage (0.884 versus 0.793). These are two within-regime tests, not a formal interaction test, and the packages

bundle memory with functional weighting. They are therefore a generalization check consistent with the fixed-output R3 interaction, not its substitute.

Descriptive external reference points. Table 7 reports three classic offline whole-piece profile-correlation analyzers from music21 on the held-out Isophonics songs.

system	coverage	exact	MIREX
frozen long package	0.884	0.732	0.782
Temperley-Kostka-Payne [8]	1.00	0.637	0.740
Krumhansl-Schmuckler [5]	1.00	0.624	0.726
Aarden-Essen [30]	1.00	0.558	0.690

Table 7: Descriptive held-out Isophonics reference points for the causal, abstaining long-memory package and three offline whole-piece profile-correlation analyzers.

The long-memory package has the highest point estimate, while Temperley-Kostka-Payne is the strongest whole-piece reference point. The submitted Krumhansl-Schmuckler contrast is +0.108 with a CI spanning zero ($[-0.008, +0.228]$, $p = 0.25$); restricting that analyzer to the detector’s claimed-event mask yields 0.625, not an independently matched-coverage operating point. No equivalence, noninferiority, or superiority test was specified. Moreover, the systems have different output forms and information: each offline analyzer reads the entire song and returns one key, whereas the causal package may change or abstain event by event. The table therefore supplies familiar descriptive context, not a claim of parity with the offline key- estimation literature.

The descriptive confusion mix is: exact 72%, fifth 7%, relative 5%, parallel 2%, other 14%. This pooled exact rate differs from the per-piece mean in Table 6; the parallel row matches the mode tilt’s development effect exactly.

Posterior calibration. Raw filtered posteriors are overconfident: on claimed exact-labeled development events, the mean top posterior is 0.929 against 0.772 exact accuracy (expected calibration error, ECE, 0.157). Overconfidence does not affect ranking, abstention, or any paired accuracy claim, but it matters the moment the probability is displayed. Post-hoc temperature scaling [31], fit once on the development split ($T = 1.55$, negative log-likelihood argmin), reduces held-out claimed-event ECE from 0.192 to 0.041. Because the transform is monotone and display-only, the held-out claims are byte-identical to the frozen artifacts, which simultaneously verifies that the calibration pass took nothing from test data.

10 Limitations

The primary R3 result is deliberately narrow: two frozen detector packages, 36 Beethoven performances, their common claimed events, and a shared 12-class diatonic-collection scoring ontology derived from 24-state outputs. It isolates the reference definition while holding input and output fixed, but it does not establish the size or direction of the effect for other

repertoires, ontologies, or detectors. R3 and R4 were declared during revision after the editorial concern was known and are exploratory; their intervals quantify uncertainty but do not convert them into preregistered confirmatory tests.

Neither reference is ground truth. Analyst-declared contexts and active notated key-signature collections encode different musical questions, and either may be debatable in ambiguous passages. The broader cross-corpus contrasts also change repertoire, observation construction, and reference provenance together. Reference persistence is one measured contributor, not a complete causal explanation.

The detector ontology contains only 12 major and 12 minor states. Modal, blues, mode-mixture, and tonic-ambiguous loops can fall outside it or admit multiple reasonable readings; three wholly modal Isophonics test tracks are consequently excluded from 24-state accuracy and coverage. They remain in the frozen split as a separate behavioral audit. The recognizer is also part of the measurement chain: fixtures embed its chord rankings, so recognizer changes require regenerating fixtures before comparison.

“Streaming” here means causal evaluation during offline replay of recorded performed MIDI. The study does not measure wall-clock latency, transport failures, interface behavior, or musician judgments in a live session. Its 30-second and 1-second packages are frozen experiment recipes, not the current application defaults; post-submission product tuning neither updates nor validates these paper-era results.

Held-out splits contain only 10 to 41 pieces, and later product work reused some of them. The offline comparison is limited to three classic profile-correlation analyzers with different information and output forms. In particular, justkeydding [10] did not build reproducibly in our environment, so no claim is made against it or newer score-based systems. Corpus licensing further prevents redistribution of two gated fixture sets; the project instead records pins, derived facts, splits, commands, and evaluation artifacts.

Finally, calibration is reference-relative just like accuracy. The display temperature in Section 9 is fit for correctness against the Isophonics region reference; it need not be calibrated against analyst contexts or another musical interpretation.

11 Conclusion

The reference construct is part of the streaming key-estimation task. On fixed Beethoven performances, detector outputs, common claimed events, and a shared 12-class diatonic-collection ontology, analyst-declared key contexts and active notated key-signature collections reverse the ranking of the same two frozen detector packages. The long-minus-short contrast changes from -0.081 under the analyst reference to $+0.080$ under the notated-signature reference, a $+0.160$ paired interaction whose interval excludes zero.

The supporting analyses delimit that result. A development grid reproduces the opposing memory effects and shows that the functional cue itself helps under When in Rome contexts and hurts under Isophonics regions. Within the overlap, longer-persistence analyst contexts progressively favor the long-memory package, but without a sharp threshold. The original held-out package reversal provides a separate generalization check while bundling two design choices and changing corpus as well as reference.

The engineering contribution is a reproducible causal selective-prediction protocol over recorded performed MIDI: coverage, accuracy on claims, change lag, and spurious switching are reported together, and the detector may abstain when evidence is insufficient. This study does not establish offline parity or end-to-end live usability. It establishes a narrower prerequisite for valid comparison: publications and benchmarks should name what their references encode, including their provenance, temporal granularity, and treatment of tonicization, signature, mode, and ambiguity.

Future work should repeat the fixed-output dual-reference design across repertoires and detector families, expand beyond the 24 major/minor states, and evaluate the causal system with musicians in live interaction. Those extensions can test whether the reversal generalizes; they should not collapse distinct reference constructs into a single undifferentiated “key accuracy.”

Competing Interests

The author has no competing interests to declare.

Reproducibility: protocol, logs, corpus pins, splits, harness, and held-out evaluation artifacts live under `research/whatkey/` in the WhatChord repository. Every number traces to a dated log entry and a report with its generating command.

<https://github.com/EarthmanMuons/whatchord>

Bibliography

- [1] G. Weber, *Theory of Musical Composition: Treated with a View to a Naturally Consecutive Arrangement of Topics*. O. Ditson & Company, 1846. [Online]. Available: <https://books.google.com/books?id=AOosAAAAYAAJ>
- [2] J. Moreno, “Subjectivity, Interpretation, and Irony in Gottfried Weber’s Analysis of Mozart’s ‘Dissonance’ Quartet,” *Music Theory Spectrum*, vol. 25, no. 1, pp. 99–120, 2003, doi: 10.1525/mts.2003.25.1.99.
- [3] R. Rowe, “Key Induction in the Context of Interactive Performance,” *Music Perception*, vol. 17, no. 4, pp. 511–530, 2000, doi: 10.2307/40285831.
- [4] P. Toiviainen and C. L. Krumhansl, “Measuring and Modeling Real-Time Responses to Music: The Dynamics of Tonality Induction,” *Perception*, vol. 32, no. 6, pp. 741–766, 2003, doi: 10.1068/p3312.
- [5] C. L. Krumhansl, *Cognitive Foundations of Musical Pitch*. Oxford University Press, 1990.
- [6] D. Temperley, “What’s Key for Key? The Krumhansl-Schmuckler Key-Finding Algorithm Reconsidered,” *Music Perception*, vol. 17, no. 1, pp. 65–100, 1999, doi: 10.2307/40285812.
- [7] J. Albrecht and D. Shanahan, “The Use of Large Corpora to Train a New Type of Key-Finding Algorithm: An Improved Treatment of the Minor Mode,” *Music Perception*, vol. 31, no. 1, pp. 59–67, 2013, doi: 10.1525/mp.2013.31.1.59.
- [8] D. Temperley, *Music and Probability*. MIT Press, 2007. doi: 10.7551/mitpress/4807.001.0001.
- [9] C. Raphael and J. Stoddard, “Functional Harmonic Analysis Using Probabilistic Models,” *Computer Music Journal*, vol. 28, no. 3, pp. 45–52, 2004, doi: 10.1162/0148926041790676.
- [10] N. Nápoles López, C. Arthur, and I. Fujinaga, “Key-Finding Based on a Hidden Markov Model and Key Profiles,” in *Proceedings of the 6th International Conference on Digital Libraries for Musicology (DLfM)*, 2019, pp. 33–37. doi: 10.1145/3358664.3358675.
- [11] G. Micchi, M. Gotham, and M. Giraud, “Not All Roads Lead to Rome: Pitch Representation and Model Architecture for Automatic Harmonic Analysis,” *Transactions of the International Society for Music Information Retrieval*, vol. 3, no. 1, pp. 42–54, 2020, doi: 10.5334/tismir.45.
- [12] N. Nápoles López, M. Gotham, and I. Fujinaga, “AugmentedNet: A Roman Numeral Analysis Network with Synthetic Training Examples and Additional Tonal Tasks,” in *Proceedings of the 22nd International Society for Music Information Retrieval Conference (ISMIR)*, 2021, pp. 404–411. doi: 10.5281/zenodo.5624533.
- [13] C. Weiß, H. Schreiber, and M. Müller, “Local Key Estimation in Music Recordings: A Case Study Across Songs, Versions, and Annotators,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 28, pp. 2919–2932, 2020, doi: 10.1109/TASLP.2020.3030485.
- [14] N. Nápoles López, L. Feisthauer, F. Levé, and I. Fujinaga, “On Local Keys, Modulations, and Tonicizations: A Dataset and Methodology for Evaluating Changes of Key,” in *Proceedings of the 7th International Conference on Digital Libraries for Musicology (DLfM)*, 2020, pp. 18–26. doi: 10.1145/3424911.3425515.
- [15] C.-H. Chuan and E. Chew, “Polyphonic Audio Key Finding Using the Spiral Array CEG Algorithm,” in *IEEE International Conference on Multimedia and Expo*, 2005, pp. 21–24. doi: 10.1109/ICME.2005.1521350.

- [16] B. J. Mizener and W. J. Dowling, “Real-Time Modulation Perception in Western Classical Music,” *Music Perception*, vol. 39, no. 5, pp. 484–502, 2022, doi: 10.1525/mp.2022.39.5.484.
- [17] M. Richards, “Tonal Ambiguity in Popular Music’s Axis Progressions,” *Music Theory Online*, vol. 23, no. 3, 2017, doi: 10.30535/mto.23.3.6.
- [18] N. J. Shea, C. W. M. White, B. Hughes, and D. T. Vuvan, “Metric Accent Affects Perception of Key Center in Pop-Music Chord Loops,” *Music Perception*, vol. 42, no. 5, pp. 467–480, 2025, doi: 10.1525/mp.2025.2321624.
- [19] T. de Clercq, “The Logic of Six-Based Minor,” *Music Theory Online*, vol. 27, no. 4, 2021, doi: 10.30535/mto.27.4.4.
- [20] “MIREX Audio Key Detection Task: Evaluation Procedures.” [Online]. Available: https://music-ir.org/mirex/wiki/2019:Audio_Key_Detection
- [21] M. Gotham, G. Micchi, N. Nápoles López, and M. Sailor, “When in Rome: A Meta-Corpus of Functional Harmonic Analysis,” *Transactions of the International Society for Music Information Retrieval*, vol. 6, no. 1, pp. 150–166, 2023, doi: 10.5334/tismir.165.
- [22] D. Tymoczko, M. Gotham, M. S. Cuthbert, and C. Ariza, “The RomanText Format: A Flexible and Standard Method for Representing Roman Numeral Analyses,” in *Proceedings of the 20th International Society for Music Information Retrieval Conference (ISMIR)*, 2019, pp. 123–129. doi: 10.5281/zenodo.3527756.
- [23] M. Mauch *et al.*, “OMRAS2 Metadata Project 2009,” in *Late-Breaking Session, 10th International Society for Music Information Retrieval Conference (ISMIR)*, 2009. [Online]. Available: <https://ismir2009.ismir.net/proceedings/LBD-18.pdf>
- [24] J. de Berardinis, A. Meroño-Peñuela, A. Poltronieri, and V. Presutti, “ChoCo: A Chord Corpus and a Data Transformation Workflow for Musical Harmony Knowledge Graphs,” *Scientific Data*, vol. 10, p. 641, 2023, doi: 10.1038/s41597-023-02410-w.
- [25] F. Foscari, A. McLeod, P. Rigaux, F. Jacquemard, and M. Sakai, “ASAP: A Dataset of Aligned Scores and Performances for Piano Transcription,” in *Proceedings of the 21st International Society for Music Information Retrieval Conference (ISMIR)*, 2020, pp. 534–541. doi: 10.5281/zenodo.4245490.
- [26] R. P. Adams and D. J. C. MacKay, “Bayesian Online Changepoint Detection,” 2007. [Online]. Available: <https://arxiv.org/abs/0710.3742>
- [27] I.-Y. Tsaknaki, F. Lillo, and P. Mazzarisi, “Bayesian Autoregressive Online Change-Point Detection with Time-Varying Parameters,” *Communications in Nonlinear Science and Numerical Simulation*, vol. 142, p. 108500, 2025, doi: 10.1016/j.cnsns.2024.108500.
- [28] R. El-Yaniv and Y. Wiener, “On the Foundations of Noise-Free Selective Classification,” *Journal of Machine Learning Research*, vol. 11, pp. 1605–1641, 2010, [Online]. Available: <https://jmlr.org/papers/v11/el-yaniv10a.html>
- [29] M. S. Cuthbert and C. Ariza, “music21: A Toolkit for Computer-Aided Musicology and Symbolic Music Data,” in *Proceedings of the 11th International Society for Music Information Retrieval Conference (ISMIR)*, 2010, pp. 637–642. doi: 10.5281/zenodo.1416114.
- [30] B. J. Aarden, “Dynamic Melodic Expectancy,” The Ohio State University, 2003. [Online]. Available: http://rave.ohiolink.edu/etdc/view?acc_num=osu1060969388
- [31] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On Calibration of Modern Neural Networks,” in *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017, pp. 1321–1330. [Online]. Available: <https://proceedings.mlr.press/v70/guo17a.html>